

Koki Madono 真殿 航輝

Assistant Professor (non-tenure-track), School of Fundamental Science and Engineering, Waseda University, Tokyo

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[Google Scholar](#) | [GitHub](#) | [Hugging Face](#) | [researchmap](#) | Bluesky: [@madorin0130.bsky.social](#)

My research interest is computer graphics for applications — supporting human creation from both computer graphics and human–computer interaction. Recent projects include data-driven ink painting rendering, interactive texture segmentation of 3D scanned models, and generative AI for robotic drawing.

Education

Apr. 2021 – Mar. 2024	Ph.D. in Engineering , Waseda University Supervisor: Edgar Simo-Serra
Apr. 2019 – Mar. 2021	M.S. , Waseda University Supervisor: Tetsuji Ogawa
Apr. 2015 – Mar. 2019	B.S. , Waseda University Supervisor: Tetsuji Ogawa

Employment

Apr. 2024 – present	Assistant Professor (non-tenure-track), School of Fundamental Science and Engineering, Waseda University Concurrent Researcher, Waseda Research Institute for Science and Engineering (2024–2026)
Aug. 2023 – present	Part-time Engineer / Internship , Preferred Networks, Tokyo Mainly mentored by Keita Higuchi
Jul. 2022 – Oct. 2022	Research Internship , Adobe Research, London (remote) Mainly mentored by Niloy J. Mitra and Jakub Fišer
May 2021 – Dec. 2021	Internship / Part-time Research Collaborator , Adobe Research, San Jose (remote) Mainly mentored by Daichi Ito, Yannick Hold-Geoffroy, and Yijun Li
Feb. 2019 – Dec. 2023	Technical Staff / Research Assistant , National Institute of Advanced Industrial Science and Technology (AIST), Tokyo Mentored by Masayuki Tanaka and Masaki Onishi

Publications

Journal

- Koki Madono**, Yuan Mingcheng, Edgar Simo-Serra. “Hint-AUC: Deterministic Region-based Hint Generation for Line Art Colorization Evaluation.” *IEEE Transactions on Visualization and Computer Graphics*, 2026 (to appear).
- Koki Madono**, Takeo Igarashi, Hiroharu Kato, Taisuke Hashimoto, Fabrice Matulic, Tsukasa Takagi, Keita Higuchi. “Interactive Texture Segmentation of 3D Scanned Models Leveraging Multiview Automatic Segmentation.” *IEEE Computer Graphics and Applications*, 2025. (also presented at CHI 2025 LBW)
- Koki Madono**, Edgar Simo-Serra. “Data-Driven Ink Painting Brushstroke Rendering.” *Computer Graphics Forum (Pacific Graphics)*, 42(7), 2023.
- Koki Madono**, Masayuki Tanaka, Masaki Onishi, Tetsuji Ogawa. “SIA-GAN: Scrambling Inversion Attack Using Generative Adversarial Network.” *IEEE Access*, 9, 2021.

Conference

- Koki Madono**, Yannick Hold-Geoffroy, Yijun Li, Daichi Ito, Jose Echevarria, Cameron Smith. “Automatic Vector Caricature via Face Parametrization.” *Pacific Graphics (Conference Proceedings)*, 2023.
- Koki Madono**, Masayuki Tanaka, Masaki Onishi. “ScrambleMix: A Privacy-Preserving Image Processing for Edge-Cloud Machine Learning.” *Pacific-Rim Symposium on Image and Video Technology (PSIVT)*, 2023.
- Koki Madono**, Masayuki Tanaka, Masaki Onishi. “Instance-wise Center Loss for Efficient Training of Deep Convolutional Neural Networks.” *IEEE Global Conference on Consumer Electronics (GCCE)*, 2022.
- Koki Madono**, Masayuki Tanaka, Masaki Onishi, Tetsuji Ogawa. “Scrambling Parameter Generation to Improve Perceptual Information Hiding.” *Electronic Imaging (HVEI)*, 2021.
- Koki Madono**, Masayuki Tanaka, Masaki Onishi, Tetsuji Ogawa. “Block-wise Scrambled Image Recognition Using Adaptation Network.” *AAAI-20 Workshop on Artificial Intelligence of Things*, 2020.
- Koki Madono**, Teppei Nakano, Tetsunori Kobayashi, Tetsuji Ogawa. “Efficient Human-In-The-Loop Object

Detection using Bi-Directional Deep SORT and Annotation-Free Segment Identification.” *APSIPA Annual Summit and Conference*, 2020.

Awards

Sep. 2024	CyberAgent Company Award, Visual Computing 2024 for “Hint-AUC: Deterministic Region-based Hint Generation for Line Art Colorization Evaluation”
Mar. 2024	Best Presentation Award, 13th Early Bird Program final report meeting, Waseda Research Institute for Science and Engineering
2022	CGVI Best Research Presentation Award (IPSJ SIG-CGVI), Visual Computing 2022 for data-driven brushstroke rendering

Competitive Funding

Apr. 2025 – Mar. 2028	JSPS KAKENHI, Grant-in-Aid for Early-Career Scientists (PI) — JPY 3,700,000 複雑な色表現を再現可能なロボット描画系列生成AIの構築 (robotic drawing-sequence generation AI reproducing complex color expression)
Oct. 2024 – Mar. 2027	JST ACT-X 「次世代AIを築く数理・情報科学の革新」 (PI) — JPY 4,500,000 完成絵を分解可能な描画系列生成AIの構築 (drawing-sequence generation AI that can decompose finished paintings)
Jun. 2023 – Mar. 2024	Early Bird Program (13th) , Waseda Research Institute for Science and Engineering — JPY 800,000
Jun. 2023 – Mar. 2024	Waseda University Grant for Special Research Projects — JPY 300,000
Oct. 2021 – Mar. 2024	JST 次世代研究者挑戦的研究プログラム (support for pioneering research initiated by the next generation) JPY 200,000/month stipend + JPY 500,000/year research funds

Patent

2023	“Vectorized Caricature Avatar Generator,” U.S. Patent Application US20230140146A1 Daichi Ito, Yijun Li, Yannick Hold-Geoffroy, Koki Madono , Jose Ignacio Echevarria Vallespi, Cameron Younger Smith
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Teaching

2024 – present	Waseda University: Science and Engineering Laboratory 1A / 1B; Introduction to Java Programming; Java Programming (Densei 1); C Programming (Densei 2)
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Academic Service

Conference	Reviewer: BMVC 2021; WACV 2024 (Rounds 1 & 2); IEEE PacificVis 2025 (Conference Track)
Journal	Reviewer: IEEE Access (2021); IEICE Transactions on Fundamentals A (2025)

Media

Jul. 2024	Featured in <i>Waseda Campus Now</i> , July 2024 issue
Apr. 2024	Waseda Weekly, 研究まっしぐら 「水墨画をきれいに描けるデジタル技術の研究」

Last updated: July 16, 2026 | madonokouki.github.io